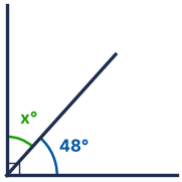
Unit 4 Classwork — Angle Relationships — Answer Key

Complementary/Supplementary · Vertical Angles · Parallel Lines · Triangle Theorems

A. Complementary, Supplementary & Linear Pair

Problem 1 Score 3

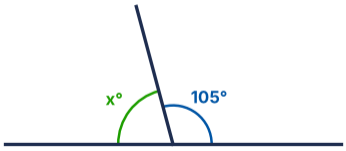
The two angles shown are complementary. Find x.



Answer: 42

Problem 2 Score 3

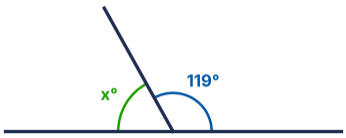
The two angles shown are supplementary. Find x.



Answer: 75

Problem 3 Score 3

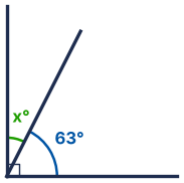
These two angles form a linear pair. Find x.



Answer: 61

Problem 4 Score 4

The two angles shown are complementary. Find x.

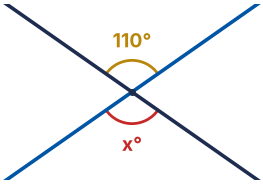


Answer: 27

B. Vertical Angles

Problem 1 Score 3

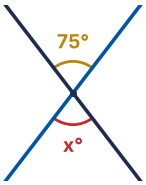
The lines shown cross at a point. Find x (vertical angles).



Answer: 110

Problem 2 Score 4

The lines shown cross at a point. Find x (vertical angles).

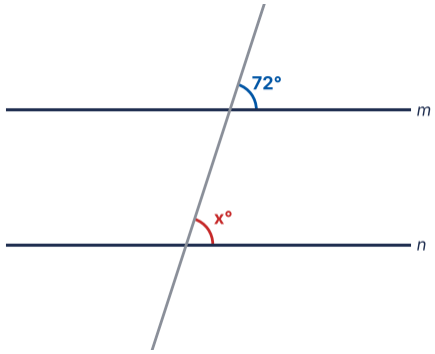


Answer: 75

C. Parallel Lines & Transversals

Problem 1 Score 3

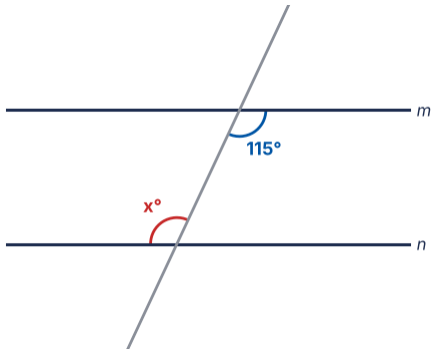
Lines m and n are parallel. The marked angles are corresponding angles. Find x.



Answer: 72

Problem 2 Score 4

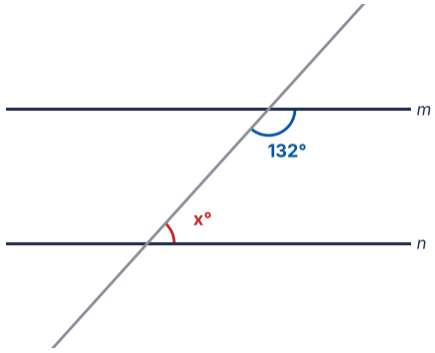
Lines m and n are parallel. The marked angles are alternate interior angles. Find x.



Answer: 115

Problem 3 Score 4

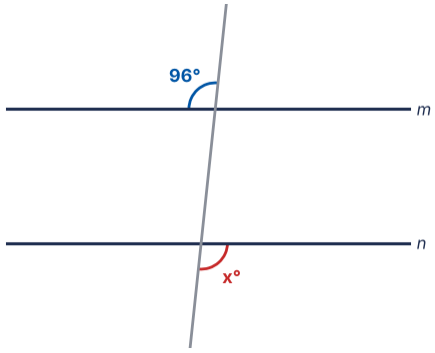
Lines m and n are parallel. The marked angles are same-side interior angles. Find x.



Answer: 48

Problem 4 Score 4

Lines m and n are parallel. The marked angles are alternate exterior angles. Find x.



Answer: 96

Problem 5 Score 5

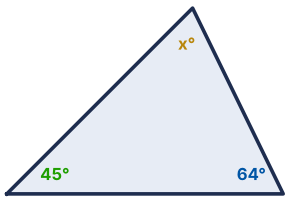
Two parallel lines are cut by a transversal. Two same-side interior angles are (3x + 20)° and (2x + 10)°. Find the LARGER angle, in degrees.

Answer: 110

D. Triangle Angle Sum & Exterior Angle Theorem

Problem 1 Score 3

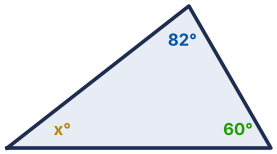
Find x in the triangle shown.



Answer: 71

Problem 2 Score 4

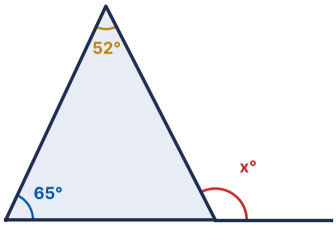
Find x in the triangle shown.



Answer: 38

Problem 3 Score 4

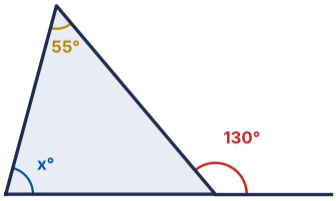
The exterior angle theorem: find x (the exterior angle).



Answer: 117

Problem 4 Score 5

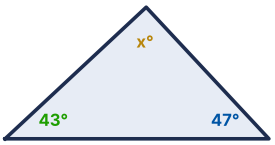
The exterior angle is 130°. One remote interior angle is 55°. Find x.



Answer: 75

Problem 5 Score 5

Find x in the triangle shown.



Answer: 90

Problem 6 Score 5

Two angles of a triangle are (2x)° and (x + 15)°. The exterior angle at the third vertex is 105°. Find the larger of those two angles, in degrees.

Answer: 60

Problem 7 Score 5

Why do the three angles of every triangle add to 180°?

A. Draw a line through the top vertex, parallel to the base. The three angles on that straight line match the triangle's three angles.

B. A triangle has 3 angles, and 3 × 60° = 180°.

C. A triangle is half of a circle, and half of 360° is 180°.

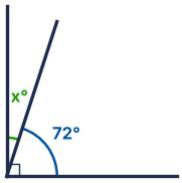
D. Every triangle has one 90° angle and two 45° angles.

Answer: A

E. Mixed Review

Problem 1 Score 4

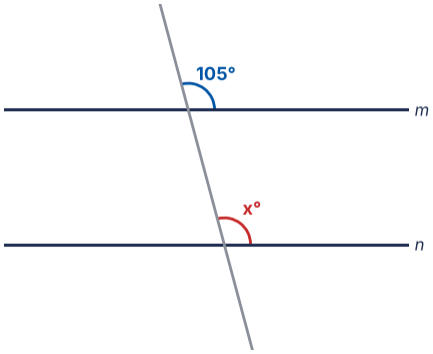
The two angles shown are complementary. Find x.



Answer: 18

Problem 2 Score 4

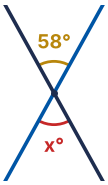
Lines m and n are parallel. The marked angles are corresponding angles. Find x.



Answer: 105

Problem 3 Score 4

The lines shown cross at a point. Find x (vertical angles).



Answer: 58